

Research Paper

Quantifying water use in New Zealand's dairy food system: A baseline for future sustainability

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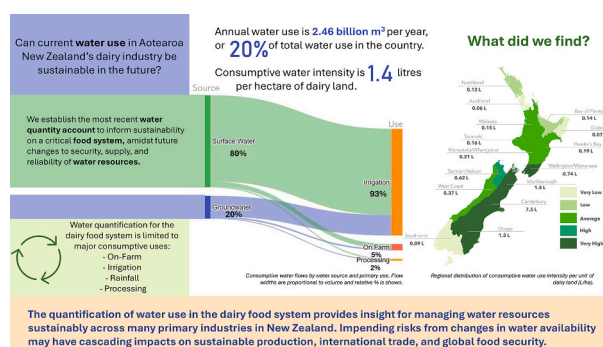
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HIGHLIGHTS

- Calculated water quantity account of the New Zealand dairy sector.
- Surface water accounts for ~80 % of national consumption.
- Water use intensity varies regionally, driven by climate and irrigation use.
- Water use for dairy makes up 20 % of New Zealand's water allocation.
- Water use must be quantified and considered for future sustainability.

GRAPHICAL ABSTRACT



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ABSTRACT

CONTEXT: In New Zealand, the dairy sector is a critical food system, contributing substantially to the nation's economy and supplying about 3 % of the world's dairy products. Despite this, a detailed assessment of water use across the sector has yet to be completed.

OBJECTIVE: This study addresses this gap by systematically quantifying blue (consumptive) and green (effective rainfall) water use in New Zealand's dairy sector up to primary processing.

METHODS: Using a regionally-specific approach, publicly available data were collected to calculate consented and actual water use for irrigation on dairy farms, and processing facilities nationwide. On-farm water use for stock drinking and milking sheds was calculated using region-specific values for irrigated and non-irrigated farms combined with the latest stock numbers. High-resolution climate data and land use data were combined to calculate green water use.

RESULTS AND CONCLUSIONS: In total, the dairy sector is responsible for approximately 20 % (2.46 billion m³ per year) of total consumptive (blue) water use in the country, over 90 % of which is driven by irrigation, and 80 % is sourced from surface water. Effective rainfall represents almost 80 % of total water use in the system (9.63 billion m³). Water use varies substantially across core dairy regions with consumptive water intensities ranging from 7.5 L/ha of dairy land in Canterbury to 0.15 L/ha in Waikato, despite both regions contributing majorly to the dairy sector.

SIGNIFICANCE: Understanding current water consumption and anticipating climate-induced changes are crucial for addressing the water supply-demand imbalance in New Zealand's dairy industry and ensuring the continued

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resilience of the food system. This work provides the most recent detailed estimate of the water quantity account, providing a platform for further analysis of the water dependence for the primary sector, comparison with other dairy-intensive countries, and more precise estimates for global studies on the water use of food systems and trade.

1. Introduction

Access to sufficient and reliable water resources is critical for New Zealand's primary food systems. Collectively, they are responsible for 74 % of allocated freshwater resources in the country (Bright et al., 2021). Within the primary sector, dairy farming is a major consumer of available freshwater, particularly for irrigation. New Zealand's dairy sector is not only important for the national economy, representing about 3.5 % of GDP and 29 % of total export goods (Ballingall and Pambudi, 2017), but also for international trade, with over 95 % of the milk produced for export, representing 3 % of world milk production (DCANZ, 2023). Traditional water resource management often treats freshwater primarily as a financial asset to meet human demands, overlooking its broader cultural, social and ecological significance. In Aotearoa, Māori regard water as a taonga (treasure) and wahi tapu (sacred place). Rooted in Māori values, Te Mana o Te Wai – a foundational concept of Aotearoa's freshwater policy – acknowledges the vital importance of protecting water for maintaining waterbodies, freshwater ecosystems and the communities that rely on them. Furthermore, freshwater is critical for providing ecosystem services that filter pollutants, control floods, cycle nutrients, sequester carbon, and support diverse habitats for invertebrates, fish, and birds, making water essential beyond its economic utility. Understanding the role of water in the dairy industry is vital for effective and sustainable use, especially given the dynamic impacts of climate change and water resource allocation.

Quantifying water use has been consistently highlighted as a significant research thrust by New Zealand government organisations (Ministry for the Environment and Stats NZ, 2023) and in the literature on New Zealand's dairy sector (Higham, 2017; Zonderland-Thomassen and Ledgard, 2012) and other major dairying countries (Sultana et al., 2014). Despite the emphasis on this water quantity gap, even at the governmental level, there is limited knowledge of how water use cascades throughout the dairy food system in New Zealand. This work directly addresses this issue by compiling the most up-to-date account of water use in the New Zealand dairy sector, including on-farm and processing life cycle stages. Understanding the water embedded in this system positions users to anticipate water availability challenges, sustainably manage water resources, and address potential inefficiencies across the food supply chain.

The total volume of water taken from New Zealand's water resources remains a knowledge gap on both a regional and national level (Ministry for the Environment, 2021). This makes it difficult to assess the sustainability of use throughout food systems and whether these systems can continue to support the status quo or if system inefficiencies need to be addressed. In a 2023 report, the Ministry for the Environment and Statistics NZ (StatsNZ) identified the uncertain resilience of the primary sector under pressures from resource use and climate change. Additionally, it noted a lack of national data on water quantity accounting (Ministry for the Environment and Stats NZ, 2023). New Zealand's freshwater environment provides not only critical habitat and usable resources but also holds cultural significance. To effectively safeguard water resources, a thorough understanding of water use is necessary.

The impacts of climate change on the primary sector are inevitable and are predicted to increasingly affect land-based primary food systems over the next century (IPCC, 2023). Consistent with human-driven climate change from increases in global greenhouse gas emissions (IPCC, 2023), New Zealand's annual average temperature has increased by 1.1 °C over the last century (StatsNZ and NIWA, 2020b). This has already been accompanied by heat waves, extreme rainfall, drought, and

flooding (Ministry for the Environment and Stats NZ, 2020). These changes are projected to continue across New Zealand, with an anticipated temperature increase between 0.7 °C and 3.0 °C by 2090 and more intense effects on the national water cycle (Mullan et al., 2018). Subsequently, this translates to changes in water resources across New Zealand, impacting seasonal and annual mean flows, low flows, and floods (Collins, 2020; Collins et al., 2021). The diminishing security of water resources, along with the imperative to halt further deterioration of the freshwater environment and operate within environmental limits, pose substantial challenges to ensuring the availability of water for New Zealand's primary sector (Ministry for Primary Industries, 2021). New Zealand's dairy systems are predominantly pasture-based, meaning future changes in water availability will impact production. The sensitivity of the primary sector to changes in future climate has been widely acknowledged, with a focus on changes to pasture growth (Ausseil et al., 2019; Clark et al., 2012; Collins et al., 2022; Harris-Wight, 2022). Pertinent to dairying regions that rely on irrigation, a predicted decline in water resource reliability and an increase in irrigation demand suggests escalating challenges for irrigation despite increases in mean river flows (Collins et al., 2022). Given the outlook on New Zealand's water availability and security, managing water resources in New Zealand requires an integration of the foreseeable impacts of climate change for water quantity allocation (New Zealand Government, 2020a). However, it is imperative to continue national assessments of water use for primary sector food systems to determine how they may be affected by future changes in water resource availability.

Climate pressures are intensifying water resource challenges across Aotearoa, with the spring of 2024 bringing a stark example: severe flooding and heavy rainfall have impacted Dunedin and Westland (Oliver and Tomsett, 2024), while Hawke's Bay and North Canterbury struggle under prolonged dry conditions (Steele, 2024; Burns, 2024). Sustainable water practices are not just crucial for New Zealand's dairy sector – they are essential to the continued health of our freshwater systems. By effectively managing water resources, it can be ensured that these systems thrive under climate uncertainty, supporting the environmental, social, cultural, and economic values that sustain our communities for future generations. The dairy sector must consider how systems and practices can be adjusted to operate within environmental limits to mitigate quantity and availability issues for the future (Ministry for Primary Industries, 2021). Ultimately, operating within these limits will position the industry to respond better to future challenges and mitigate adverse impacts to enable long-term sustainability. To achieve this vision and enable effective change, it is imperative to first gain an understanding of water use dependence throughout the system.

Investigations of water embedded in food systems have been performed with the water footprint concept (Hoekstra, 2017) and have since been applied to many large agricultural regions like the United States (Dang et al., 2015), India (Harris et al., 2020), and Southern Africa (Conway et al., 2015), and to the trade of global food commodities (D'Odorico et al., 2019; Konar et al., 2011). Although agriculture, particularly the dairy sector, is a major industry in New Zealand, only the energy embedded within the system has been examined in detail (Fitzgerald et al., 2016; Norton, 2018). Few studies have contributed to a comprehensive system analysis of water use across the sector. Water use for New Zealand pastoral dairy farms has been quantified at a farm scale (Higham, 2017), and a water scarcity footprint has been determined for an irrigated and a non-irrigated dairy region in New Zealand (Payen et al., 2018). Regional water use for pasture irrigation and stock purposes in New Zealand between 1997 and 2000 was quantified

(Flemmer and Flemmer, 2007). However, this work was based only on allocated water use, and no updated or more complete water quantity accounting assessments have been done since. Assessments of climate change impacts on water resource availability and irrigation water demand across the country have also been completed (Collins et al., 2022), and New Zealand's National Science Challenges have worked on investigating the impacts on land use suitability and the hydrological cycle under future climate change (Ausseil et al., 2019; Collins et al., 2021). While the influence of climate change on pasture growth and potential adaptation options for the dairy sector has been well studied in New Zealand (Lee et al., 2012; Lee et al., 2013), less is understood about the broader impacts on the sector, particularly for water resources. Despite the increasing awareness of the criticality of water use in the food system, a quantitative investigation of the water requirements for the entire life cycle of New Zealand's largest agricultural sector remains a significant knowledge gap. Notably, a detailed understanding of water use across the dairy system and the consequences of this use under increasing freshwater challenges is lacking.

This research has two primary objectives. This research has two primary objectives. First, a systems-level, quantitative account of embedded water inputs in New Zealand's dairy industry, tracking water flows from source to primary use. This informs better decision-making for future sustainable and equitable management of water resources by determining where the water is used and where it is sourced. This outcome also supports the research community by contributing to the growing body of knowledge surrounding the water use of global food systems. Second, an increased understanding of the inherent risks regarding water use within the current system in the context of future changes to the security, supply and reliability of water resources. This will address a critical knowledge gap by connecting availability and demand for a key industry within the primary sector. Existing research has investigated the magnitude and timing of potential climate change effects on irrigation and agriculture in New Zealand; however, the impact on specific food systems, such as dairy, is limited. Consequently, achieving these objectives will deepen our understanding of the New Zealand dairy industry's dependence on water resources, particularly to

identify the impacts that predicted changes in water resource availability and reliability may have on water use in the sector.

2. Methods

This work establishes the most recent water accounting analysis of the New Zealand dairy system. This covers systems accounting for total consumptive water use (blue water) for dairy production and processing, corresponding to an estimated 19.0 % of New Zealand's nationally allocated freshwater resources. In this work, water consumption refers to the water used directly for on-farm dairy production and off-farm processing. Water quantification for the dairy food system is performed on a regional scale across New Zealand. This comprises major consumptive water use that is typically associated with the dairy industry, such as irrigation, stock water, milking shed water use, and dairy processing water. Indirect water use, including the water consumed in producing bought-in feed, seeds, pesticides, fertilisers, fuel, electricity, and packaging, has been excluded. This work also examines annual effective rainfall volumes for dairy land in New Zealand. Water use has been quantified for each component annually for 2018–2022, with an average value used to represent the current status quo of the industry. Fig. 1 shows a descriptive process diagram of the data inputs and general methodology in this analysis.

Water consumption for each component of the food system is categorised by water source (surface water – including storage and dam supply, groundwater, and precipitation) and is limited to freshwater only. Where necessary, this work relies on conservative assumptions that may overestimate total volumes attributed to the dairy system. Water takes for permitted activities, such as for stock water, or takes below regional council restriction levels (variable depending on the water body and environmental conditions) for volume and rate are not included.

2.1. Irrigation

Freshwater use for irrigation was estimated using both consented

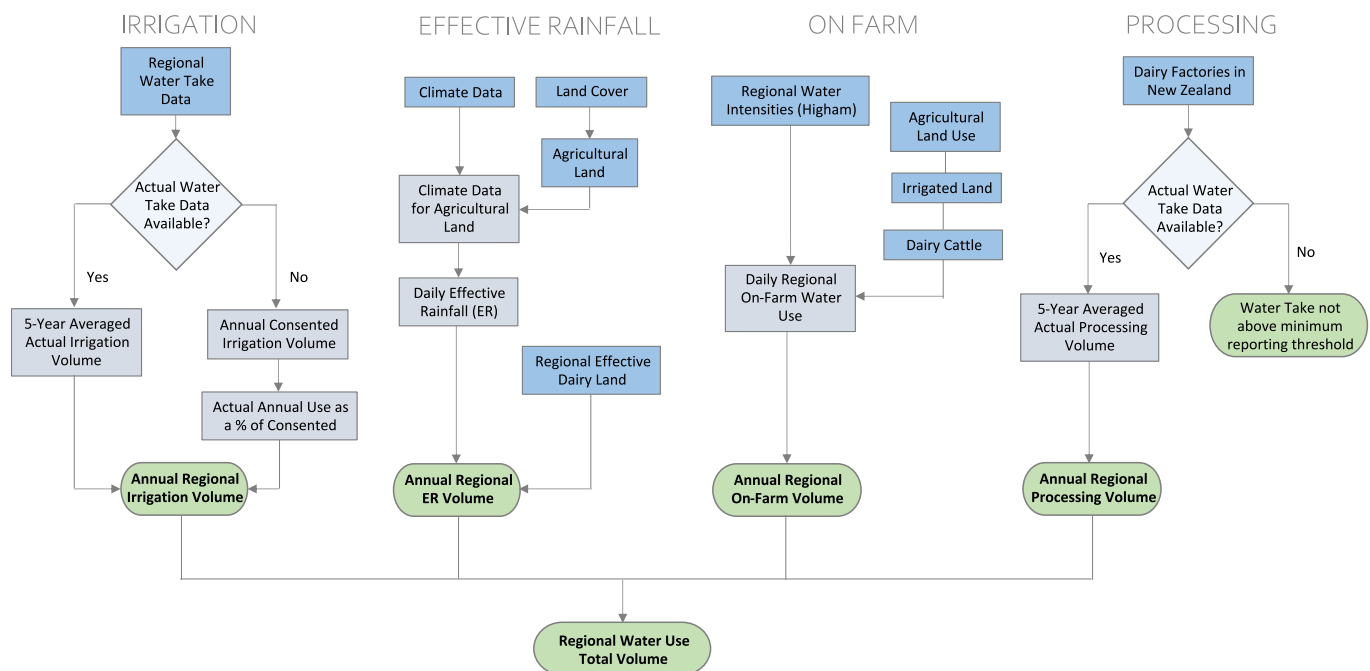


Fig. 1. Process diagram for the water quantity accounting approach used in this study for the New Zealand dairy food system. The system boundary includes consumptive (blue) water use (irrigation, on farm, and processing) and estimates of effective rainfall (green water). Blue boxes indicate data inputs, and grey boxes indicate calculations and/or data manipulations. Green boxes represent the final calculated volumes for system components and the total system. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and actual water use data. These data were obtained from the 16 regional council bodies in New Zealand for all active consents, as well as expired consents that could continue to abstract water under S124 of the Resource Management Act (New Zealand Government, 1991).

In New Zealand, the management of consent information for water takes varies across regional councils. These differences include the classification of consented land use and water use and the categorisation of consent numbers and their associated Water Abstraction Points (WAP). This resulted in a dual approach to estimating water for irrigation: if available, actual water use data were selected for dairy consents and aggregated annually, if not, a proportion of the consented water use volumes were used as a proxy. For regions in which consented volumes were used to estimate irrigation volumes, where annual volumes were not explicitly included, the annual volume for irrigation takes was calculated by multiplying instantaneous flow (in L/s) over a typical 16-week irrigation season. This was informed by accepted approximations for annual volumes (Rajanayaka et al., 2010). In New Zealand, the amount of water that can be taken for a consent is strictly regulated, so water take consents offer a valuable source of data on 'allowable' water use volumes. However, we note there is often a gap between this 'allowable' volume and actual metered water use.

Water resource consents often serve multiple purposes, with many permitting water use for activities beyond those listed in the 'Primary Use' category. For example, a consent may have pasture irrigation recorded as its primary use, but water is also consented for stock purposes or municipal supply. While including all consents associated with pasture irrigation may lead to a potential overestimation of water use, differentiating water volumes for specific uses within each consent is not feasible. Nonetheless, considering the significant contribution of irrigation to water use, any overestimation is relatively minimal. Furthermore, recent amendments to the National Resource Management (Measurement and Reporting of Water Takes) Regulations 2010 require that daily water use data must be reported to regional councils for all water takes greater than 20 L/s (this condition will be reduced to >5 L/s by September 2026) (New Zealand Government, 2020b). Excluding smaller irrigation takes has negligible impact on water quantity accounting accuracy, as their contribution to total water use in New Zealand is minimal.

There is an inherent risk in using recorded water volumes as it can be difficult to distinguish anomalies from actual values. Water use data may contain gaps, spikes, and other erroneous values that do not accurately reflect actual water use. Council data is often raw and may lack quality control, resulting in missing data caused by a variety of factors (faulty meters, weather damage, manual submission errors). Erroneous values, particularly where records were greater than three times the consented volume, were replaced with the consented volume. Since the data was verified against consented volumes, this method may still account for actual abstractions that exceed allocations.

2.2. Effective rainfall

Daily climate data (2017–2022) from the National Institute of Water and Atmospheric Research's (NIWA) Virtual Climate Station Network (VCSN) was used to determine the effective rainfall (ER) used for pasture growth by considering gross rainfall, potential evapotranspiration (PET), soil moisture, and runoff. VCSN is a climate database producing estimates of climate variables interpolated across a regular ~5 km grid of New Zealand.

The New Zealand Land Cover Database (LCDB) was used to isolate VCSN nodes that were within 1 km of agricultural land. Agricultural land was assumed to be incorporated into the 'Short Rotation Cropland', 'High Producing Exotic Grassland', and 'Low Producing Grassland' classes identified in the LCDB. In total, 7730 VCSN nodes were selected to represent the climate across dairy-producing regions in New Zealand (equivalent to about 67 % of the VCSN database).

Within the VCSN, soil moisture is derived using daily rainfall and

PET, assuming rainfall infiltrates into a soil store from which moisture is depleted by evapotranspiration loss and surface runoff occurs when rainfall exceeds the saturated soil store capacity. Effective rainfall (mm) was calculated for each region by aggregating the daily rainfall volumes for days in which the soil moisture was less than zero from the VCSN nodes within the region. Once the soil moisture is greater than zero, it was assumed that the excess rainfall became surface runoff. ER volumes for each region were estimated using the annual ER (mm) and multiplying by the effective land area used for dairy, obtained from DairyNZ (Dairy, 2021b).

2.3. On-farm water

In this study, on-farm water use excludes irrigation, as it is a major component that has been assessed separately. There is limited information on water use for pastoral dairy farms, as most of the published values are thought to be outdated or are industry averages based on historical data for the entire country (Higham, 2017). Recent research, however, has begun to address this data gap by implementing detailed measurements of water use on irrigated and non-irrigated farms throughout New Zealand, developing predictive models for agricultural water use accounting (Higham, 2017). Specifically, dairy farm water use has been modelled regionally for irrigated and non-irrigated dairy farms between 2010 and 2015 to estimate average water use (L/cow per day) for "stockwater" (drinking water for stock, SDW) and "milking parlour water" (water used in the milking shed, MPW). Higham's regionally modelled average water use for SDW and MPW has since been adopted by industry bodies (Dairy, 2021a) and has been used for the on-farm water use quantification in this work.

Irrigated land (StatsNZ, 2021b) and agricultural land use (StatsNZ, 2021a) area statistics from StatsNZ were used to determine the percentage of land used for dairy in each region. Values from the more complete dataset (2017) were used in this study. However, Auckland and Gisborne regional data possessed high sample errors or imputation levels and were suppressed by StatsNZ. Since irrigation is not common in these regions and they represent a small proportion of the New Zealand dairy industry, all farms are assumed to be 'non-irrigated' to mitigate the missing data. Total effective farm area (hectares) and regional cattle numbers were available from Dairy NZ annual statistics (Dairy, 2021b). Region classifications across these multiple datasets differed slightly, and adjustments were made where appropriate (See SI Section S2 for further details). It was assumed that there is a proportionate distribution of dairy cattle across both irrigated and non-irrigated dairy farms in New Zealand, such that water use could be directly calculated from total dairy cattle numbers.

For each region, on-farm water use volumes were estimated for irrigated (*i*) and non-irrigated (*n*) dairy farms using the following equation defining the on-farm water use for each region, *r*, and use (SDW or MPW), *u*, as $OFW_{r,u}$. C_r is the regional number of dairy cows, *LUF* is the regional irrigated land use fraction, and *WI* is the water intensity in cubic metres per cow.

$$OFW_{r,u} = C_r \times [(WI_{r,u,i} \times LUF_{r,i}) + (WI_{r,u,n} \times LUF_{r,n})]$$

Estimating the water quantities for on-farm water use from individual council consents is challenging not only because collecting, cleaning, and aggregating the data is time-consuming but also because the reported data likely does not accurately represent water use. In many cases, consent rates are low, and consent holders are not required to report actual water use. Further, on-farm water consents can include water use for other purposes which may be impossible to distinguish from on-farm use. The source of on-farm water was estimated by assigning the same regional water source proportion (i.e., groundwater or surface water) of all consumptive water takes in the region from the StatsNZ consented freshwater takes dataset (StatsNZ and NIWA, 2020a).

2.4. Dairy processing

Dairy processing facilities receive milk directly from farm suppliers to produce and export products such as pasteurised milk, cream, butter, casein, whey and milk powders. Dairy manufacturing facilities receive products from dairy processors to produce value-added dairy products, including cheese, yoghurt, and ice cream. This research focuses primarily on the dairy processors that receive milk from the farm gate, as it represents the immediate stage following on-farm production in the dairy food supply chain. By establishing the research boundary at this point, a comprehensive analysis can be conducted, encompassing dairy processing and export.

Several dairy processing plants receive milk directly from farm suppliers, but their processing operation is small-scale and does not possess water take consents within their region. Either the water takes for these factories are considered within the permitted activity water take volumes specified by the regional council and don't require a consent, or they are drawing water from the reticulated municipal supply. If this is the case, they are not required to report water use volumes to the council and would be difficult to ascertain in this work. This exclusion is justified due to their relatively low water consumption, which is likely to have a minor impact on total volumes and freshwater resources.

Water is used in several stages of dairy manufacturing, including in sanitisation, clean-in-place (CIP) process, heating, and cooling, and it can be supplied from multiple sources depending on the quality required. Water consumption can also vary significantly across different dairy processing facilities, depending on product mix and water efficiency. Where possible, annual water use volumes were obtained directly from dairy processors for the period between 2017 and 2022. For the remaining factories, volumes were obtained from regional councils.

3. Results and discussion

The dairy sector is responsible for approximately 19 % (2.46 billion m^3 per year) of total New Zealand freshwater consumption, with the

largest overall consumers being Canterbury and Otago (Fig. 2). Consumptive water use for New Zealand's dairy system is dominated by irrigation, particularly in the South Island (93.1 %). Regions characterised by high irrigation use relative to effective rainfall, namely Canterbury, Otago, Marlborough, and Wellington/Wairarapa, exhibit the highest levels of intensive water consumption when weighted by regional dairy land and cow numbers. On average, 2.28 billion m^3 (92.6 %) of all water used by New Zealand dairy farms was used for irrigation. Total water consumption for stock purposes was 137 million m^3 (5.6 %), and for dairy processing was 45.0 million m^3 (1.8 %). Effective rainfall is also a critical input for dairying across all regions in New Zealand and represents 79.6 % (9.63 billion m^3) of total water use for New Zealand dairying. Thus, the total water quantity account for the New Zealand dairy sector is 12.1 billion m^3 .

The dairy industry in New Zealand is hinged on the security, reliability, and availability of both groundwater and surface water resources. Approximately 80.1 % of consumptive water use for New Zealand dairy is sourced from surface water. This trend is consistent across both islands, with surface water being the source of consumptive water use for 73.6 % and 80.6 % of dairying in the North Island and South Island, respectively. Waikato, Taranaki, and Marlborough are most dependent on surface water, and Wellington/Wairarapa, Bay of Plenty and Hawke's Bay are most dependent on groundwater resources.

Before this investigation, the most up-to-date study on annual water use for irrigation and stock purposes for New Zealand dairy farms assessed 14 regions (excluding the West Coast) between 1997/98 and 2000/01 (Flemmer and Flemmer, 2007). Over the last two decades, water use for the dairy food system across New Zealand has intensified with a 59 % increase (from 1.017 billion m^3) in the estimated blue water use (consumptive use) and a stronger reliance on surface water resources (an 8.1 % increase). Our estimates in this study correspond to a 61 % increase (from 892 million m^3) in irrigation water use, representing an increasing proportion of total consumptive water use from 88 % to 93 % over the last 20 years. Notably, there was only an 8.8 % increase (from 125 million m^3) in water use for stock purposes, despite a nearly 30 % increase in dairy cow numbers. This suggests that water use efficiency for stock purposes, although small relative to other system components,

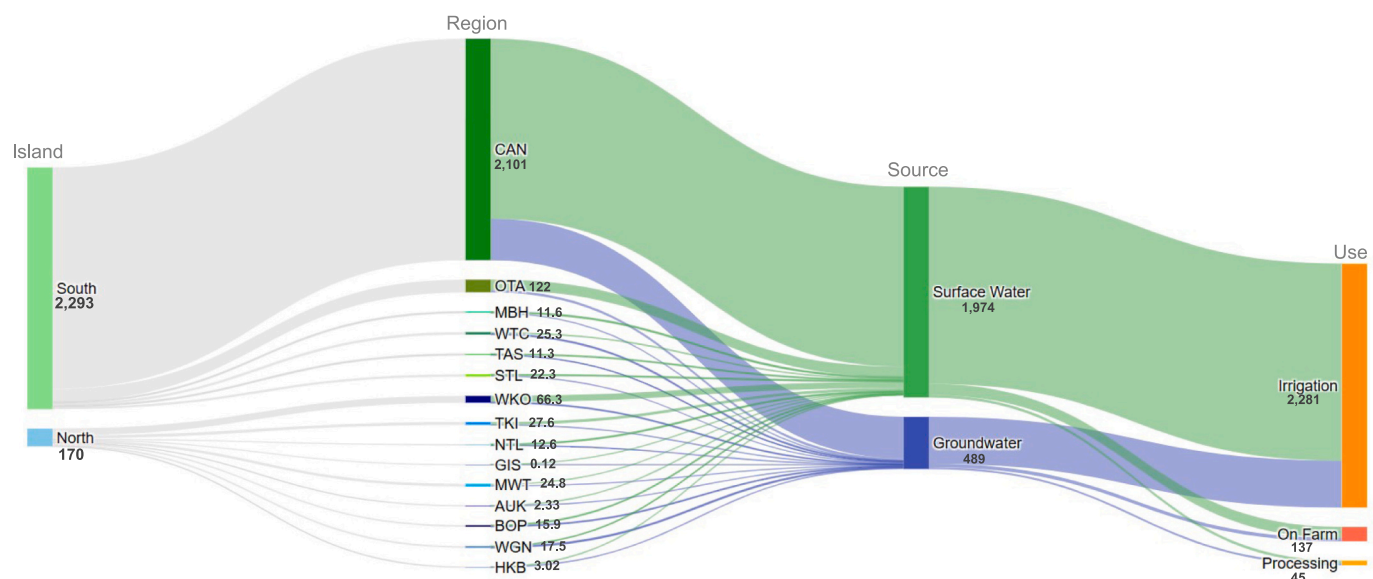


Fig. 2. Consumptive water flows (excluding effective rainfall) through the dairy food system in New Zealand by island, region, water source, and primary use. Flow widths are proportional to volume and sum to 2.46 billion m^3 vertically across the figure. Coloured flows represent the proportion of surface water (green) and groundwater (blue) used in each region, across the country, and in each life cycle stage studied. Regions are abbreviated by their region code: Northland: NTL, Auckland: AUK, Waikato: WKO, Bay of Plenty: BOP, Gisborne: GIS, Hawke's Bay: HKB, Taranaki: TKI, Manawatu-Whanganui: MWT, Wellington/Wairarapa: WLG, Tasman/Nelson: TAS, West Coast: WTC, Marlborough: MBH, Canterbury: CAN, Otago, OTA, and Southland: STL. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

has improved over the last 20 years. Similar to the results in this study, Canterbury and Otago were highlighted as the largest water-consuming regions, with 62.4 % and 15.8 % of the national total, respectively.

3.1. Regional water use

Regionally, water use quantification for dairying included consumptive water volumes for irrigation, on-farm use, and processing, in addition to effective rainfall (Fig. 3). Regional variations in dairying systems in New Zealand are influenced by factors such as climate, soil, and access to processing facilities, resulting in unique water utilisation patterns within regional dairy industries. Understanding these regional differences is crucial for developing effective water management strategies tailored to the needs of each region's farming practices and to its future predicted changes in water availability and security. Total regional water use ranges from 13.2 million m³ in Gisborne to 3.8 billion m³ in Canterbury (including ER), as shown in Fig. 3. Total consumptive regional water use ranges from 0.12 million m³ in Gisborne to 2.1 billion m³ in Canterbury (excluding ER). This water use can be separated into the different components of the dairy system, namely irrigation (Fig. 3A), effective rainfall (Fig. 3B), on-farm use (Fig. 3C), and processing (Fig. 3E).

Dairy farms make up approximately 60.5 % of all irrigated area in New Zealand (StatsNZ, 2021b). Water use for irrigation of dairy pasture is dominated by use in the Canterbury region, with 89.9 % (corresponding to 2.05 billion m³) of the national total, followed by Otago (112.9 million m³). Regional intricacies in climate, irrigation practices, and water management approaches influence water use, with some regions heavily reliant on irrigation to support dairy farming while others benefit from higher rainfall and less irrigation demand. Although effective rainfall accounts for about 80 % of water use on a national scale in Aotearoa's dairy system, there remains a strong reliance on irrigation in certain regions. Seasonal variability often misaligns rainfall with peak farm water demands, while location-specific water requirements mean that rainfall alone is insufficient for consistent pasture growth. For example, irrigation provides necessary and consistent water supply for pasture growth in Canterbury and has facilitated the growth of the dairy industry in the region (Pangborn and Woodford, 2011). Without irrigation in Canterbury, water availability would be significantly limited, making it challenging to sustain dairy farming operations. Irrigation in Canterbury is primarily facilitated through large-scale irrigation schemes that use water from rivers, lakes, and groundwater sources and is regulated within allocation zones (Environment Canterbury, 2009). Alternatively, Southland experiences relatively high rainfall compared to other regions, and irrigation is less common. However, some dairy farms continue to utilise irrigation, mainly during dry periods, to ensure consistent pasture growth and quality.

Effective rainfall and on-farm water volumes tend to reflect the prevalence of dairying in each region, as water use is estimated directly from regional effective dairy land or cow numbers. As a result, the regions with the greatest area of land used for dairy (see SI Section S1) – Waikato, Canterbury, Southland, and Taranaki – exhibit the largest effective rainfall and on-farm water use volumes. Conversely, in regions where dairy farming is less common – Gisborne, Marlborough, and Hawke's Bay – these volumes are much lower.

The regional consumptive water use intensity for dairy varies across the country from 0.07 L/ha of dairy land in Gisborne to 7.5 L/ha of dairy land in Canterbury (Fig. 4). Water use intensity is quantified as the volume of water consumed within the dairy system per hectare of effective dairy land, providing a measure of the water consumption efficiency within the regional industry. This consumptive water use intensity excludes ER (i.e., green water), indicating the direct impact on surface and groundwater resources. Nonetheless, ER has an appreciable impact on regional water use profiles, potentially reducing required irrigation volumes in regions with high ER (see SI Section S3, Fig. S3.1). However, regions with high ER may still heavily depend on water

resources to combat drought and support pasture growth, as this rainfall may not occur when needed the most.

The distribution of dairy processing facilities is clustered in the main dairy farming regions in New Zealand, guaranteeing proximity to milk supply. Most of the factories are in Waikato and Canterbury (see SI Section S1), corresponding to the largest volumes for processing water use with 13.1 million m³ and 11.8 million m³ of annual processing water consumption, respectively. Water is required for several different processes within these facilities, including clean-in-place for milk dryers and their components, truck flushing, steam generation in boilers, cooling water, and adding ingredients to produce a slurry for drying. Every processing facility is different, so water consumption within each facility will be driven by their product mix and water use efficiency/recovery systems. Moreover, this may change year-on-year, so individual factories may exhibit significant fluctuations in water use depending on what is being produced. In this work, we assume these fluctuations have been captured within the range of uncertainty for each region.

To gain a comprehensive understanding of the impact of the dairy food system on regional water resources, we examine the consumptive water use specifically attributed to dairying as a proportion of the region's total allocated freshwater. For regions in which the dairy industry comprises a significant proportion of allocated freshwater (Fig. 5), including Canterbury (34 %), Tasman (16 %), Taranaki (15 %), Southland (11 %), West Coast (9 %), and Waikato (9 %), changes to water availability and reliability are likely to have a significant impact on the dairy system. Alternatively, in regions where consumptive water use for dairy represents a small proportion of allocated freshwater (<5 %), there are likely to be other food systems that rely on water resources in the region, resulting in unique pressures for water supply and demand. However, understanding these pressures remains relevant, as it is likely that the impacts on the dairy sector would also be present for other food systems. For example, although consumptive water use for dairying in Marlborough is relatively high (1.5 L/ha), dairying only represents 2 % of the total allocated freshwater. Understanding water resource challenges and the potential impacts on the dairy system in all regions can provide valuable insights for managing water resources sustainably across many primary industries.

3.2. Climate

The global water cycle is increasingly unstable – with consequences for the world's economy, populations, and ecosystems. This is the core message of a report by the Global Commission on the Economics of Water, which calls for reframing water as a global common good and embedding sustainable water use across all sectors (Global Commission on the Economics of Water, 2024). For Aotearoa, this means prioritising efficient and sustainable management of water to reduce climate risks and safeguard long-term environmental and societal outcomes. Furthermore, it is critical to anticipate the timing and magnitude of changes to water resources from climate change. Maintaining an up-to-date understanding of current water use, as achieved in this work, is an essential element of addressing this challenge. Here, we further qualitatively examine the implications of water consumption in New Zealand's dairy industry, considering future predicted changes to climate, water availability, and resource reliability.

Predicted increases in drought and greater variability in river flows will have implications for both rain-fed and irrigated dairy farming, as well as water resource management. Generally, there is a pattern of increased precipitation in the west and south of New Zealand and decreases in the north and east. Soil conditions are projected to become drier in the spring and summer and wetter in the winter, with more prolonged dry periods in the northern and eastern North Island, Tasman, Marlborough, and western Canterbury, but wetter for the West Coast and parts of eastern Canterbury (Mullan et al., 2018). Low river flow conditions are expected to worsen and occur sooner after the wet season across many regions, except for the West Coast. River flow reliability –

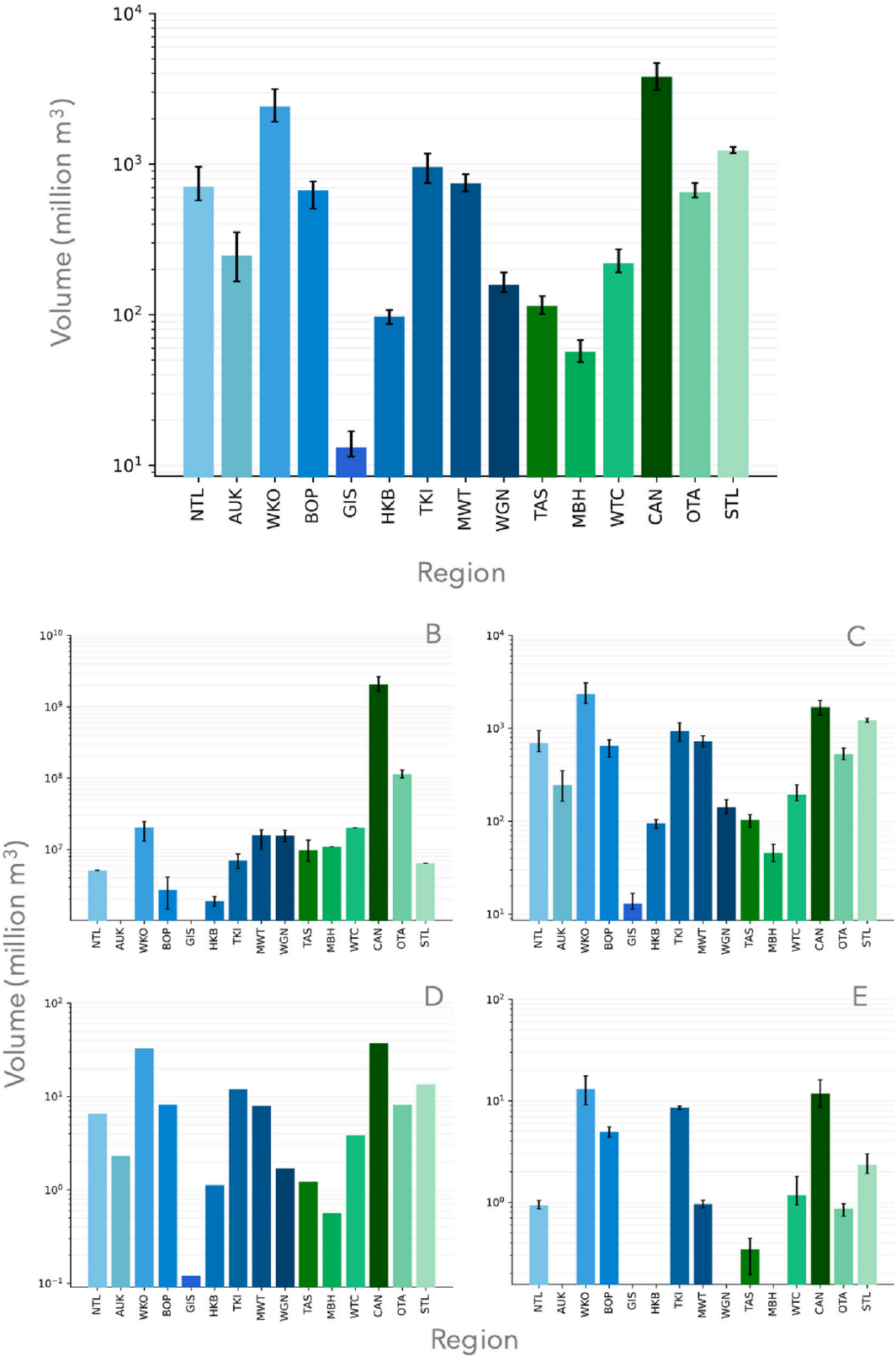


Fig. 3. Average total (A), irrigation (B), effective rainfall (C), on-farm (D), and processing (E) water use in the New Zealand Dairy sector. Regions are shown on the x-axes, ordered from Northern-most to Southern-most and coloured by island (blue = North Island, green = South Island). Error bars for each plot and region represent the maximum and minimum values across the 5-year study range. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

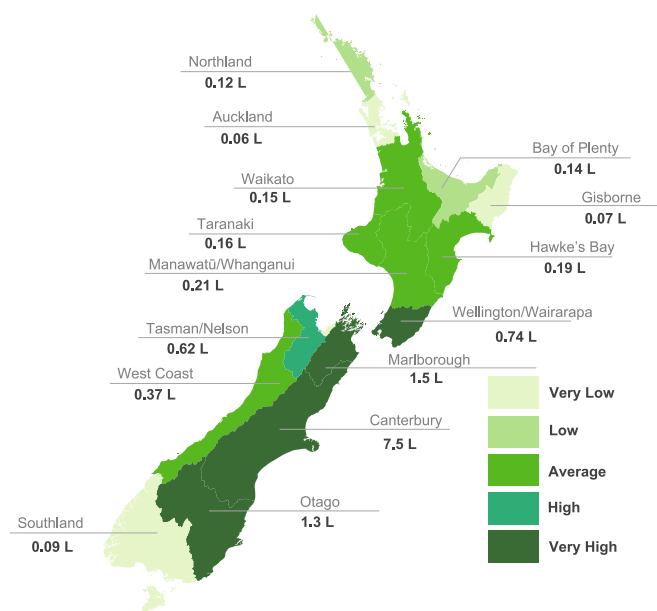


Fig. 4. Regional distribution of consumptive water (i.e., blue water) use intensity per unit of dairy land (L/ha), where the deepest green represents the largest water use intensity for dairy, mostly concentrated on the Eastern side of the South Island. Intensities are logarithmically classified into Very Low (0–0.1 L/ha), Low (0.1–0.15 L/ha), Average (0.15–0.6 L/ha), High (0.6–0.7 L/ha), and Very High (0.7+ L/ha). The overall national consumptive water intensity is 1.4 L/ha. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

the proportion of time water may be abstracted from a river under low flow conditions – will generally decline across New Zealand. Changes in river flows are more variable for the South Island, resulting in a variable

supply of irrigation water (Collins et al., 2021), which is particularly relevant for the irrigation-dependent regions of Canterbury and Otago. Under these projections, it is realistic to expect ensuing increases in irrigation demand to maintain the same agricultural yield. Marked increases in irrigation water demand are projected across most of New Zealand, with effects on the South Island emerging later in the century than in the North Island (Collins et al., 2022). It is important to note that while in some areas, seasonal water availability increases, the decline in short-term supply reliability and increases in crop water needs reinforce the projected imbalance of irrigation supply and demand. To maintain dairy production, more water is likely to be necessary in the future.

The regional water use intensity illustrates the relative water use efficiency of the dairy sector in different dairying regions (Fig. 4). For example, the Waikato region has one of the lowest consumptive water intensities (0.15 L/ha of dairy land) in the country despite being the largest dairying region (approximately 25 % of total dairy cows and dairy land) due to its very low reliance on irrigation with just over 3 % of total dairy land being irrigated. Conversely, within the Canterbury region, which is the second most influential dairy region, dairy land is almost entirely irrigated (97.5 %), and consequently, this region has the highest consumptive water use intensity. While these values represent two extremes in New Zealand dairy water use practices, they also highlight the need for region-specific solutions to ensure future sustainable water use. Taking an “umbrella” approach to an issue that varies substantially across regions is problematic. This is because national projections lack usefulness at regional and local levels. Effective management of water resources requires accounting for the unique characteristics of each region, including climate, geography, water resource availability, and community factors. To plan for the long-term sustainable use of water resources, further research is needed to understand how to manage the long-term effects of climate change on New Zealand’s pasture-based dairy systems and identify actions to enhance sector resilience. Assessing the extent and timing of changes in water supply and demand at a regional level on a sub-annual timescale is

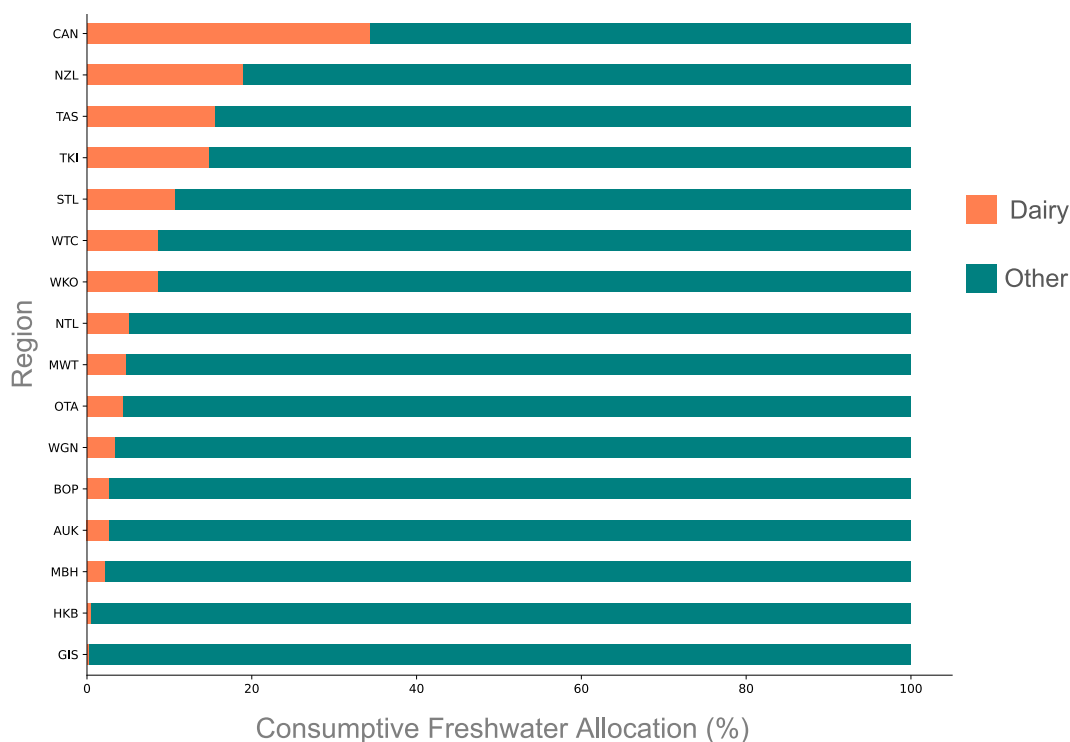


Fig. 5. Relative proportions of regional consented freshwater takes (excluding non-consumptive and hydroelectric use) used for dairying (StatsNZ and NIWA, 2020a). Regions are in descending order, including the proportion for the whole of New Zealand (NZL). Consented freshwater takes for dairy in most regions are less than 5 % of the regional allocated total, except for Canterbury (CAN), Tasman (TAS), Taranaki (TKI), Southland (STL), West Coast (WTC), and Waikato (WKO).

critical for informed decision-making and effective management of shifting water availability.

The regions heavily reliant on surface water (Northland, Waikato, Taranaki, Marlborough, Canterbury, and Otago) are more likely to be directly impacted by climate change and the greater variability of surface water flows (Fig. 6). However, groundwater resources will also likely be impacted by lower and less frequent recharge rates, as well as changes to irrigation demand. In some regions, excessive reliance on groundwater resources is already a concern, as water use is already approaching allocation limits (e.g., in Canterbury). Further, there is limited accurate regional information about the volume of groundwater stores, the rates of recharge, and the factors influencing recharge that could be impacted by climate change (Collins et al., 2012). This gap poses a substantial challenge in determining the long-term sustainable use of groundwater resources.

The dairy food system faces impending risks from shifts in water availability and security, given water is a key input for the sector. This has cascading effects on sustainable production, international trade, and global food security. Sustainable water resource management in Aotearoa's dairy system is critical to balancing economic viability and environmental sustainability. Understanding water use secures farm productivity, supports local economies, and protects freshwater ecosystems – ensuring resilience in a changing climate. Currently, the sector accounts for around 20 % of New Zealand's exports, which accounts for approximately 3 % of world dairy production (Ballingall and Pambudi, 2017). NZ is the world's 8th largest milk producer, exporting over 95 % of the milk produced onshore to more than 130 countries worldwide, including China, Australia, Indonesia, United States, and Japan (DCANZ, 2023). This means that 3 % of the global dairy system and many regions worldwide are reliant on New Zealand's water resources.

3.3. Data limitations

Similar to other water accounting studies, data availability and quality were limiting factors for this analysis. Firstly, each region has different approaches to monitoring its water resources, with varying levels of publicly available information. Many regional councils host

open data portals where water resource consent information can be downloaded, while some regions require a council data request for consents. Further to this, water use in New Zealand is mostly governed via water consents, whereby a user is consented to use a specific volume or rate from a given surface water or groundwater source. Often, the actual water used for a given application differs from the consented volume. Although there are national regulations for the measurement and reporting of water takes, freshwater resources are managed by regional authorities, which have an array of different processes and frameworks for collecting and managing water use information. This means that water use data is not reported uniformly by all regional councils, nor is it easily accessible, requiring data requests to councils. Even in cases where both consented and actual water use data were obtained, the quality was variable. There is currently no mechanism to verify the accuracy of reported water use volumes, so these values are assumed to be a complete account. In this analysis, both the consented and actual water use volumes reported are relevant for estimating irrigation and processing water use.

Given this work is the first account of its kind in New Zealand, this analysis used a conservative approach, aligning with practices from other research in systems analysis for energy (Grubert and Sanders, 2018) and food (Marston et al., 2018) systems. For example, in cases where actual data were missing, consented volumes were used instead, likely resulting in an overestimation of actual water use. This was the case for irrigation water use for three of the 16 regional councils (Nelson, Marlborough, and West Coast), where no actual water use data was available. In cases where actual water use data were not reported for dairy processing facilities, these facilities were assumed to be below reporting thresholds and, therefore, have a minimal influence on the total water use calculation. To account for variable data quality, errors in reporting, and undetected anomalies, five years of water use volumes were collected and averaged for all available regions for both irrigation and processing water use.

On-farm water use data for the dairy sector were not available at the individual farm level and are impossible to parse from reported consented or actual water use data (i.e., only total on-farm water use is reported, which often includes water used for non-dairy livestock and

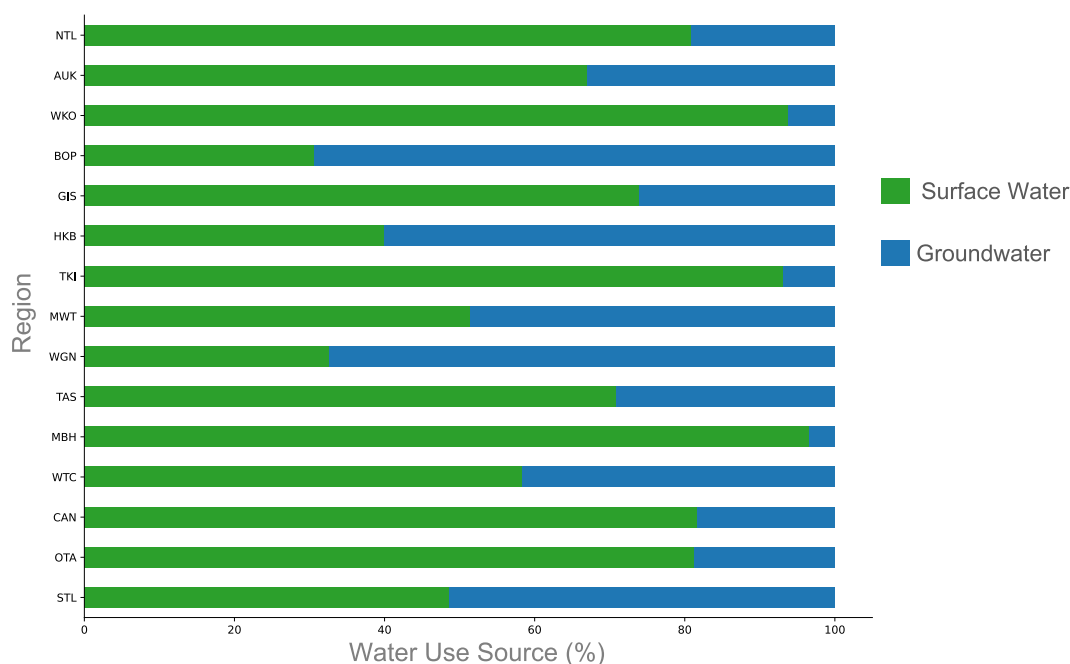


Fig. 6. Relative proportions of regional groundwater (blue) and surface water (green) use for the dairy sector. Regions are ordered from Northern-most (top) to Southern-most (bottom). Most regions show a higher reliance (>50 %) on surface water resources, except for the Bay of Plenty (BOP), Hawke's Bay (HKB), and Wellington/Wairarapa (WGN). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

other agriculture). However, regionally specific on-farm water use intensities are available in the literature to address this challenge (Higham, 2017).

This type of systems analysis often relies on data reported from national, regional, or local governments or organisations. This presents an obvious challenge for data monitoring and quality if standards are inconsistent or missing entirely. As noted by many other studies (e.g., Chini and Stillwell, 2016), communication with individual monitoring organisations is often necessary, and data collection and aggregation occur over long timeframes. Despite this challenge, we believe that this account contributes to the quantitative rigour of estimates for water use for the dairy sector and water quantity accounting in New Zealand. Notably, the need for more accurate accounts of and standards for water use has been highlighted by the National Policy Statement on Freshwater Management (New Zealand Government, 2020a) and has been highlighted as a significant research gap at the national level (Ministry for the Environment and Stats NZ, 2023). We provide here a detailed database of water use estimates and intensities that can inform further research in the country as well as international research on water use in the dairy sector, contributing to the growing body of work on water use for traded food commodities.

4. Conclusions

This study presents the most recent estimates of the blue (consumptive) and green (effective rainfall) water quantity account of the New Zealand dairy system. Specifically, we provide a systems-level analysis of the embedded water inputs across the primary food chain, tracking water flows regionally from source to primary use, and we assess the risk to water resource use in the context of future changes in availability and security. This is pertinent in informing Aotearoa's freshwater and resource management policy, given the cultural, ecological, and environmental significance of freshwater and its role in governing the economic viability of all primary industries.

The dairy food system in New Zealand accounts for about 20 % of total national allocated freshwater consumption. The total water use for the New Zealand dairy sector is 12.1 billion m³ per year, which includes 2.46 billion m³ of blue water and 9.63 billion m³ of green water. Of the blue water quantity, 2.28 billion m³ (92.6 %) was used for irrigation, 137 million m³ (5.6 %) was used for stock purposes, and 45.0 million m³ (1.8 %) was used for dairy processing. Canterbury and Otago are the largest water-consuming regions, with the South Island representing 93.1 % of the total blue water use. Surface water is a major source of water for the dairy industry across New Zealand (approximately 80.1 % of all consumptive use), with most regions having a greater reliance (>50 %) on surface water resources than groundwater, except for Bay of Plenty, Hawke's Bay, and Wellington/Wairarapa. Regional water use intensity represents the water consumption efficiency within regional dairy industries and ranges from 0.07 L/ha of dairy land in Gisborne to 7.5 L/ha of dairy land in Canterbury.

The use of water resources is vulnerable to shifts in precipitation patterns, river flows, and irrigation demand for all of New Zealand's dairy system, regardless of whether the regional dairy industry is heavily irrigated (e.g., Canterbury and Otago) or has greater green water use. For regions in which the dairy industry comprises a significant proportion of allocated freshwater, including Canterbury (34 %), Tasman (16 %), Taranaki (15 %), Southland (11 %), West Coast (9 %), and Waikato (9 %), changes to water availability and reliability are likely to have a significant impact on the dairy system. Regions heavily reliant on surface water (e.g., Northland, Waikato, Taranaki, Marlborough, Canterbury, and Otago) face direct impacts from climate change and variable surface water flows. Groundwater resources are also vulnerable to these pressures; however, there is a lack of knowledge of the extent and timing of these effects. Given that New Zealand's dairy industry forms a critical component of the national economy, as well as the global export market, changes in water resource availability and reliability are expected to

have a profound cascading impact.

With this work, we address a critical knowledge gap in the quantification of water resource use for key primary industries in New Zealand. Further, we provide a detailed account and discussion of the process of sourcing, cleaning, and using publicly available data to inform water quantity accounting. Finally, we contribute to the published body of knowledge on the water quantity accounts of food systems worldwide with this up-to-date estimate for one of the world's key dairy-producing regions.

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CRediT authorship contribution statement

Nicole A. Cameron: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rebecca A.M. Peer:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors have nothing to declare.

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Appendix A. Supplementary data

Supplementary information to this article can be found online at <https://doi.org/10.1016/j.agry.2025.104272>.

Data availability

All data from this study is available on request to the corresponding author.

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